- CALCOLO SCIENTIFICO.

INTRODUZIONE

(ISCRIVETEVI A MOODLE: CALCOLO SCIENTIFICO 2026)

* EQUAZIONI DIFFERENZIALI

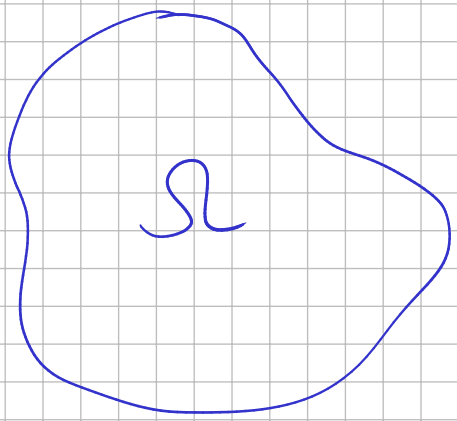
ALLE DERIVATE PARZIALI

FISICA → MODELLI MATEMATICI

- PARTIAL DIFFERENTIAL EQUATION

ANALISI NUMERICA

$\boxed{\text{APPROSSIMA SOLUZIONI PDE}}$ ← $\boxed{\text{SOLUZIONI MATEMATICHE}}$ (ESATTE)



$\Omega \subset \mathbb{R}^d \qquad d = 1, 2, 3$

Ψ

x

$t \in \mathbb{R}^+$

$u : \Omega \to \mathbb{R}^s \qquad s \in \mathbb{N}$

ORDINE $k \in \mathbb{N}$

$$F(\nabla^{(k)} u, \nabla^{(k-1)} u, \dots \nabla u, u) = 0$$

$\boxed{s = 1}$

$u : \Omega \to \mathbb{R}$

→ u ✓ → $\nabla u = \begin{pmatrix} \partial_x u \\ \partial_y u \end{pmatrix}$

2D $\quad \Omega \subseteq \mathbb{R}^2 \quad (x, y) \in \Omega \subseteq \mathbb{R}^2$

$$\nabla^{(2)} u = \begin{pmatrix} \partial_{xx} u & \partial_{xy} u \\ \partial_{yx} u & \partial_{yy} u \end{pmatrix}$$

u SUFF REGOLARE

$$\partial_{xy} u \overset{?}{=} \partial_{yx} u$$

$$\nabla^{(3)} u = \left(\begin{pmatrix} \partial_{xxx} u & \partial_{xxy} u \\ \partial_{xyx} u & \partial_{xyy} u \end{pmatrix}, \begin{pmatrix} \partial_{yxx} u & \partial_{yxy} u \\ \partial_{yyx} u & \partial_{yyy} u \end{pmatrix} \right)$$

$u \in \mathbb{R}$ $\quad \nabla u \in \mathbb{R}^d$ $\quad \nabla^{(2)} u \in \mathbb{R}^{d \times d}$ $\quad \nabla^{(3)} u \in \mathbb{R}^{d \times d \times d}$

- TRASPORTO $\quad \partial_t u + c\, \partial_x u = 0 \qquad c \in \mathbb{R}$
- CALORE $\quad \partial_t u - c\, \partial_{xx} u = 0$
- BURGERS $\quad \partial_t u + u\, \partial_x u = 0$
- LAPLACE $\quad -(\partial_{xx} u + \partial_{yy} u) = \cancel{0}\ f$
- NAVIER-STOKES INCOMPRIBILE

$s = 1 + d$

$u \in \mathbb{R}^d$

$p \in \mathbb{R}$

$$\partial_t u - \underbrace{\nu \Delta u}_{\text{VISCOSITÀ}} + \overbrace{u \cdot \nabla u}^{\text{"TRASPORTO"}} + \nabla p = f$$

$$-\operatorname{div}(u) = 0$$

$$u \cdot \nabla u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}^T \cdot \begin{pmatrix} \partial_x u_1 & \partial_y u_1 \\ \partial_x u_2 & \partial_y u_2 \end{pmatrix}^T$$

$$= \begin{bmatrix} u_1 \partial_x u_1 + u_1 \partial_x u_2 \\ + u_2 \partial_y u_1 + u_2 \partial_y u_2 \end{bmatrix}$$

$$\operatorname{div}(u) := \sum_{i=1}^{d} \partial_{x_i} u_i = \partial_x u_1 + \partial_y u_2$$

$$\operatorname{div} : \mathbb{R}^d \to \mathbb{R}$$

- COSA VOLETE FARE L'ANNO PROSSIMO?

NOTAZIONE

$u(t, x, y, \dots)$ $\qquad \partial \neq d$

$$\partial_t u(t, \dots) = \frac{\partial}{\partial t} u(t, \dots) = u_t(t, \dots)$$

$$\frac{d}{dt} u(t, x(t), y(t), \dots) = \partial_t u(t, \dots) + \partial_x u(t, x(t), \dots) \cdot \frac{\partial x}{\partial t} + \partial_y u(\dots) \cdot \frac{\partial y}{\partial t} + \dots$$

$$\nabla u = \begin{pmatrix} \partial_x u \\ \partial_y u \end{pmatrix} \in \mathbb{R}^2 \quad \left(\underbrace{u(x, y) : \mathbb{R}^2 \to \mathbb{R}}_{} \right)$$

$$\Delta u = \partial_{xx} u + \partial_{yy} u = \partial_x^2 u + \partial_y^2 u$$

$$\partial_x^2 u = \partial_x \partial_x u = \partial_{xx} u$$

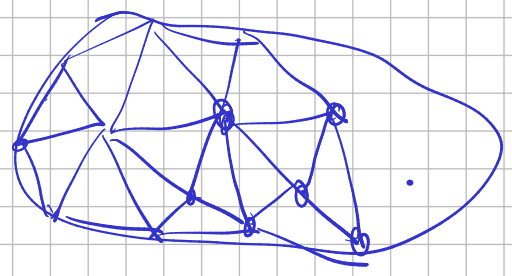
$$\operatorname{div}\left(\begin{pmatrix} u \\ v \end{pmatrix}\right) = \nabla \cdot \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \partial_x \\ \partial_y \end{pmatrix} \cdot \begin{pmatrix} u \\ v \end{pmatrix} = \partial_x u + \partial_y v$$

$$\nabla \times \begin{pmatrix} u \\ v \end{pmatrix} = \underline{\partial_x v - \partial_y u} \qquad (\nabla \times) : \mathbb{R}^2 \to \mathbb{R}$$

$$\nabla \times \begin{pmatrix} u \\ v \\ w \end{pmatrix} = \begin{pmatrix} \partial_y w - \partial_z v \\ \partial_z u - \partial_x w \\ \partial_x v - \partial_y u \end{pmatrix} = \det \begin{pmatrix} \underline{i} & \partial_x & u \\ \underline{j} & \partial_y & v \\ \underline{k} & \partial_z & w \end{pmatrix}$$

• METODI NUMERICI APPROSSIMARE SOL PDE

1.

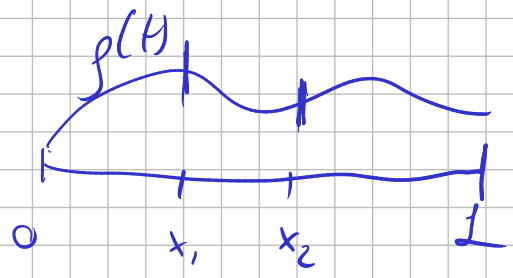


2. APPROSSIMARE u IN SPAZIO FINITO DIMENSIONALE

3. DISCRETIZZAZIONE DELLA PDE

4. TROVARE LA SOLUZIONE

MOTIVATION



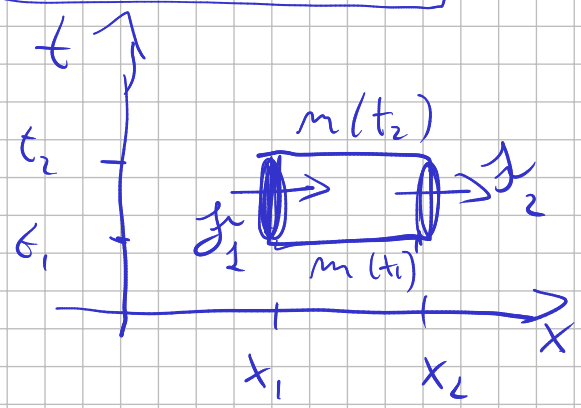
$\Omega = [0,1] \subset \mathbb{R}$

ρ DENSITÀ $\quad \rho : \Omega \to \mathbb{R}^+$

$$m_{[x_1,x_2]}(t) := \int_{x_1}^{x_2} \rho(x,t)\, dx$$

$\partial_t \rho = ?$

$$F(t,x_1) = \underbrace{\rho(t,x_1)\cdot v(t,x_1)}$$



$$\underbrace{\int_{x_1}^{x_2} \rho(t_2,x)\,dx}_{m_{[x_1,x_2]}(t_2)} - \underbrace{\int_{x_1}^{x_2} \rho(t_1,x)\,dx}_{m_{[x_1,x_2]}(t_1)} = \ast \qquad \forall\, x_1, x_2,\ t_1, t_2$$

$$\ast = \underbrace{\int_{t_1}^{t_2} \rho(t,x_1)\cdot v(t,x_1)\,dt}_{\mathcal{F}_1} - \underbrace{\int_{t_1}^{t_2} \rho(t,x_2)\cdot v(t,x_2)\,dt}_{\mathcal{F}_2}$$

FLUSSO DA SX $\qquad$ FLUSSO A DX

- DERIVIAMO IN t_2

$$\int_{x_1}^{x_2} \partial_t \rho(t,x)\,dx = \rho(t,x_1)\,v(t,x_1) - \rho(t,x_2)\cdot v(t,x_2)$$

- DERIVO IN x_2

$$\partial_t \rho(t,x) = -\partial_x\big(\rho(t,x)\cdot v(t,x)\big)$$

$$\partial_t \rho(t,x) + \partial_x\big(\rho(t,x)\cdot v(t,x)\big) = 0$$

$$\partial_t \rho(t,x) + \partial_x\big(f(t,x,\rho)\big) = 0 \qquad \forall\, x, t$$

PDE STAZIONARIA $\qquad$ ORDINE K $\qquad \partial_t u = 0$

$$F(x, u, Du, \ldots D^{(K)}u, \underbrace{g}_{\text{DATO/SORGENTE}}) = 0$$

$$\begin{pmatrix} \partial_x u \\ \partial_y u \end{pmatrix}$$

K $\qquad p_1, \quad , p_d \; : \; p_1 + \dots + p_d \leq K$

$x_1, \dots, x_d$

$$\frac{\partial^{p_1 + \dots + p_d} u}{\partial^{p_1}_{x_1} \cdots \partial^{p_d}_{x_d}}$$

- NON STAZIONARIA $\qquad$ DIPENDE ANCHE DAL TEMPO

$$F\left(x, t, \partial_t u, Du, \partial_t Du, \frac{\partial^{p_0 + \dots + p_d} u}{\partial^{p_0}_t \partial^{p_1}_{x_1} \cdots \partial^{p_d}_{x_d}}, g\right) = 0$$

$p_0 + \dots + p_d \leq K$

- SOLUZIONE CLASSICA DI UNA PDE (NON STAZ)

$$u \in \mathcal{C}^K(\Omega \times \underset{t}{[0,T]})$$